

Data analytics for personalized genomics and precision medicine

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Overview of genome sequencing

Map the genome and count the fragments/short reads of genes.

Then try to reconstruct those genes based on the principle that overlapping region in genes are connected sequentially (not physically) when not divided.

The length of short read is about 200bp in Illumina sequencing machine.

Problem when doing reconstruction:

- One base overlaps another base is unknown due to being identically the same.

Example: (AAA) and (AAT) can be reconstructed as (AAAT) or (AAAAT)

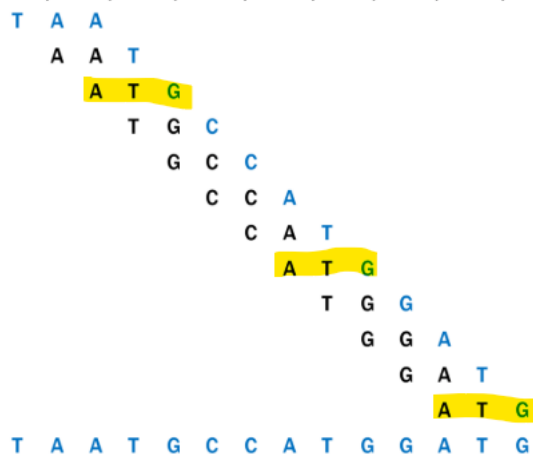
- Finding the copy number of a sequence.

Example: it might look like (AATAAATAATAAT... (AAT)x60).

- The identical short reads can be inserted into multiple regions.

Example: the base (ATG) indicated with yellow highlight.

❖TAA, AAT, ATG, TGC, GCC, CCA, CAT, ATG, TGG, GGA, GAT, ATG



Solution: Increase the equipment strength, i.e. use better equipment to get more base pair reads. Try to compare the maximum similarity in those reads. For example: to construct the sequence with the sequence with 3 bases. A maximum of 2 overlaps is needed to reduce the false positive chances. Reduce the factor of 4 for each overlapping base.

Genome mapping

Slide each read along the genome and calculate the difference using dynamic programming, until there is a read with no difference, the target gene is found.

Problem:

Some identical genes overlap with the target gene region.

Solution:

Make assumptions that the signals are noise or simply delete the entry based on different situations.

Data pre-processing

Examples of data cleaning problems:

- Noise
- Missing data
- Duplicate data
- Unnormalized data
- Categorical data

Noise definition: modification of original value

Solution: finding the average value between nearby sample points and average out the signal.

Outliers definition: data objects with characteristics that are considerably different than most of the other data objects in the data set.

Solution: Ignore the outliers

Missing value reason: consenting issues or not applicable to some categories.

Solution:

- eliminate data objects
- Estimate missing values
- Ignore the missing values during analysis
- Replace all the possible value with weighted probabilities

Duplicate data reason: merging heterogeneous data (data from totally different sources)

Problem: causing overfitting/ skewing/ biased distribution in dataset.

Solution: Delete the data object

Unnormalized data definition: measuring the correlation between attributes with different level of measurement.

Solution: Min-Max normalization (treating the distribution as uniform)/Z-score normalization (treating the distribution as Gaussian distribution)

Categorical data problem: The computer cannot store 'Male' as a value.

Solution: Use one-hot encoding, treat Male as a category and assign binary numbers under the category.

Overall data cleaning problems and solutions

	Definition	Problem	Solution
Noise	modification of original value	N/A	finding the average value between nearby sample points and average out the signal.
Outliers	data objects with characteristics that are considerably different than most of the other data objects in the data set.	Affecting the location statistics of data distribution (filled in by guessing the missing idea/'value' since there is none on the slide)	Ignore the outliers
Missing data	N/A	No data entry due to consenting issues or not applicable to some categories.	1.) Eliminate data objects 2.) Estimate missing values 3.) Ignore the missing values during analysis 4.) Replace all possible value with weighted probabilities
Duplicate data	N/A	merging heterogenous data (data from totally different sources), causing overfitting/ skewing/ biased distribution in dataset.	Delete the data object
Unnormalized data	measuring the correlation between attributes with different levels of measurement.	Heavily influenced by one attribute instead of both attributes, cannot find the true correlation between attributes	1.) Min-Max normalization (treating the distribution as uniform)

			2.) Z-score normalization (treating the distribution as Gaussian distribution)
Categorical data	N/A	The computer cannot store 'Male' as a value.	Use one-hot encoding, treat 'Male' as a category and assign binary numbers under the category.

The workflow of data cleaning:

Denoise the data if necessary → remove outliers → Handling missing data → Remove duplicate data → Categorical data encoding → Data normalization

Note that the order for data cleaning **should not be changed**. For example, if a 'true' data A has the same data as B due to noise. If there is no denoising of the data and removed data point A for being duplicated. The 'true' data point is falsely voided and distribution obtained does not reflect the real data distribution.

Summary statistics

Definition: summary statistics are numbers that summarize properties of the data.

Example: frequency, location and spread. (mode, mean, variance)

For location: mean and median are commonly used since mean alone is prone to outliers.

For spread: Min, Max, Variance are commonly used, is sensitive to outliers, median absolute valued and interquartile range is used as alternatives.

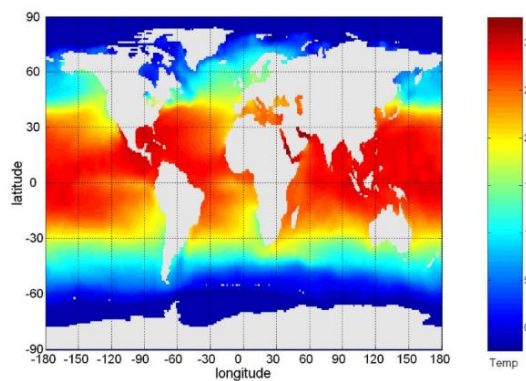
For frequency: percentage and mode are commonly used.

Percentile: a % value is used for finding data **less than/equal or less than** other data points.

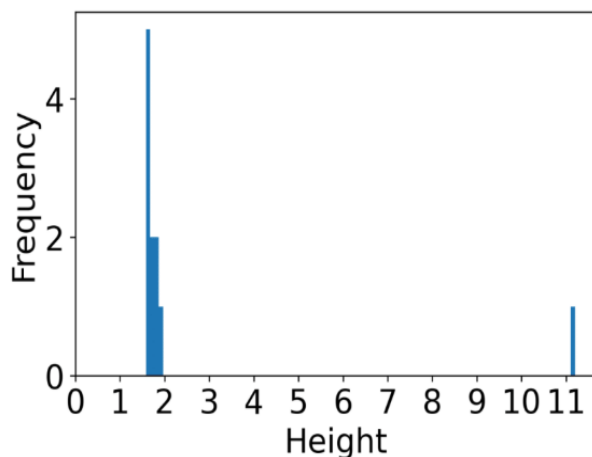
Visualization

Visualization is the conversion of data into a visual or tabular format so that the characteristics of the data and the relationships among data items can be analyzed and reported.

It can be used to identify general patterns and trends and outliers and unusual patterns. (data quality)



From this picture, you can find trends from this graph made from triplets (temperature vs longitude) or (temperature vs latitude)). Like the smaller the magnitude of the latitude, the hotter the temperature.



This histogram can identify the outlier at height 11 based on our vision instead of complex mathematical tools.

Useful links or resources:

<https://learning-oreilly-com.easyaccess2.lib.cuhk.edu.hk/library/view/python-for-data/9781449323592/>

(Online book)

https://julac-cuhk.primo.exlibrisgroup.com/discovery/fulldisplay?docid=alma991031787559703407&context=L&vid=852JULAC_CUHK:CUHK&lang=en&search_scope=All&adaptor=Local%20Search%20Engine&isFrbr=true&tab=default_tab&query=any,contains,%20python%20for%20data%20analysis&sortby=date_d&facet=frbrgroupid,include,9059592172205317984&offset=0 (physical book status in CUHK library)

Python for Data Analysis

Chapter 4: NumPy Basics: Arrays and Vectorized Computation

Chapter 5: Getting Started with Pandas

Chapter 7: Data Wrangling: Clean, Transform, Merge, Reshape

<https://github.com/VGP/vgp-assembly/tree/master/pipeline> (a tool for genome assemblies for vertebrates)

